

PARTS LIST

Semiconductors:

- IC1 - PIC18F4685 microcontroller
- IC2 - ENC28J60 Ethernet controller
- IC3 - 25LC256 EEPROM
- IC4 - 74HCT125 quadruple 3-state buffer
- IC5 - MAX232 dual driver/receiver
- IC6 - 7805, 5V regulator
- IC7 - LM3940, 5V to 3.3V converter
- LCD1 - 16x2 LCD display
- LED1-LED4 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 330-ohm
- R2 - 4.7-kilo-ohm
- R3 - 470-ohm
- R4 - 10-kilo-ohm
- R5-R7, R9-R10 - 220-ohm
- R8 - 2.7-kilo-ohm
- R11-R14 - 49.9-ohm
- R15 - 100-ohm
- VR1 - 10-kilo-ohm preset

Capacitors:

- C1 - 0.33 μ F ceramic
- C2, C5, C12-C16 - 0.1 μ F ceramic
- C3 - 0.47 μ F ceramic
- C4 - 33 μ F, 10V electrolytic
- C6-C7 - 33pF ceramic
- C8-C9 - 18pF ceramic
- C10 - 10 μ F, 10V electrolytic
- C11 - 0.01 μ F ceramic

Miscellaneous:

- X_{osc1} - 10MHz crystal oscillator
- X_{osc2} - 25MHz crystal oscillator
- CON1 - 2-pin terminal block
- CON2 - DB9 female COM port connector
- CON3 - RJ45 Ethernet connector
- CON4-CON5 - 2-pin connector

06:18 AM,29,116,http://cdn.worldweatheronline.net/images/symbols01_png_64/wsymb0002_sunny_intervals.png,PartlyCloudy,10,16,37,NE,0,0,60,10,1015,40 2014-01-14,30,86,21,70,11,17,37,NE,116,http://cdn.worldweatheronline.net/images/symbols

01_png_64/wsymb0002_sunny_intervals.png,PartlyCloudy,0.1 2014-01-15,31,88,21,70,9,14,30,NN E,116,http://cdn.worldweatheronline.net/images/symbols01_png_64/wsymb0002_sunny_intervals.png,PartlyCloudy,0.1

The client software

The TCP/IP client board, which houses Microchip TCP/IP Stack 5.4.2, is configured to work as client. More detail on the stack can be obtained from www.microchip.com website.

The general TCP Client.c file available in stack is suitably modified to carry out the following tasks:



Fig. 2: Block diagram of the weather forecast monitoring system

TABLE II
Test Points

Test point	Details
TP0	0V
TP1	9V
TP2	5V
TP3	3.3V
TP4	Train of pulses
TP5	Train of pulses

1. Send required URL to the server
2. Remove the HTML header from the incoming weather data from the server
3. Store weather data in the EEPROM
4. Send the stored data from the EEPROM to the host PC via COM port

Circuit and working

The block diagram of weather forecast monitoring system is shown in Fig. 2 and circuit diagram of the client board is shown in Fig. 3. The circuit contains Microchip PIC18F4685 (IC1), Ethernet controller ENC28J60 (IC2), EEPROM 25LC256 (IC3), quadruple 3-state buffer 74HCT125 (IC4), dual driver/receiver MAX232 (IC5), 16x2 LCD display and magnetic jack RJ45 Ethernet connector.

Ethernet controller ENC28J60 operates with 3.3V supply, so level-shifter IC 74HCT125 is used to maintain proper TTL levels between PIC and ENC via SPI bus. LCD is wired in a 4-bit mode with data lines connected to port pins RD0 through RD3 of the microcontroller IC1. The control lines RS, R/ $\overline{W}$ and EN are connected to RD4, RD5 and RA5. Status LED2 connected to RA2 is used to indicate working of the stack by blinking.

Pins RC3, RC4 and RC5 are used for SPI communication. Pins RC6 and RC7 are used as TX and RX lines for serial communication. Dual driver/receiver MAX232 provides necessary voltage levels for serial communication

between the PC and the client board. RB3 and RB4 pins act as $\overline{CS}$ lines for the

ENC and EEPROM, respectively. Pin RB5 acts as reset line for ENC28J60. EEPROM 25LC256 is used for storing incoming weather data.

The client board should be connected to the Internet through Ethernet port of the modem or router.

Power supply. The client board circuit is powered by 5V and 3.3V supplies (refer Fig. 4). A 9V DC supply is fed to the 5V regulator 7805 (IC6). This 5V supply drives LM3940 (IC7) and the other circuit including PIC18F4685 microcontroller, LCD display and level shifter 74HCT125. LM3940 is a 5V to 3.3V converter. ENC28J60 is powered by the 3.3V supply. LED1 is used to indicate the presence of power in the circuit.

Host software

The host software is developed using Delphi language, which is simple to use and does not require DLL files like VB for running its application exe. The Delphi exe file can run on any windows platform. Delphi front panel view of the weather monitor is shown in Fig. 5.

The Setting button at the screen's bottom is used to configure the COM port parameters such as port number and baud rate. The Open button opens the relevant COM port, sends City Name in the edit box window and character 'S' to the client board which initiates the start of the client program. The Show button is enabled only after all the processing is over; it enables the user to view the data in the Report List. If a query is sent for a city whose weather data is not available, the server sends 'Unavailable' message to the client. That information is passed on to the user by displaying the message through a pop-up window.

Free third-party COM port library, SDL component Report List library and Scroll Message library are used,